EE 316

Section 3

Pre Lab 10

By: Hannah Kelley

Small Signal Amplification with JFETs

Introduction:

This lab explores small-signal AC amplification, frequency response, and transistor dynamic behavior in field-effect transistor circuits. By evaluating a common-source JFET amplifier via simulation and hardware, the experiment demonstrates how external capacitors limit low-frequency performance while internal parasitic capacitances bound high-frequency bandwidth. The exercise teaches how to construct Bode magnitude plots to identify midband gain, cutoff frequencies, and bandwidth, while illustrating signal phase inversion. Additionally, it highlights the practical disparities between idealized simulation models and real-world hardware limits.

Simulations and Results:

The JFET amplifier circuit in Figure 1 was built and simulated in LTSpice. Figure 2 shows a screenshot of the JFET amplifier circuit in LTSpice. The circuit was simulated over a range of frequencies from 30 Hz to 16 MHz. Figure 3 shows the Bode plot for this simulation. The bandwidth and cutoff frequencies for the Bode plot in Figure 3 are noted in Table 1. Table 2 shows selected V_{out} and gain values from the Bode plot in Figure 3.

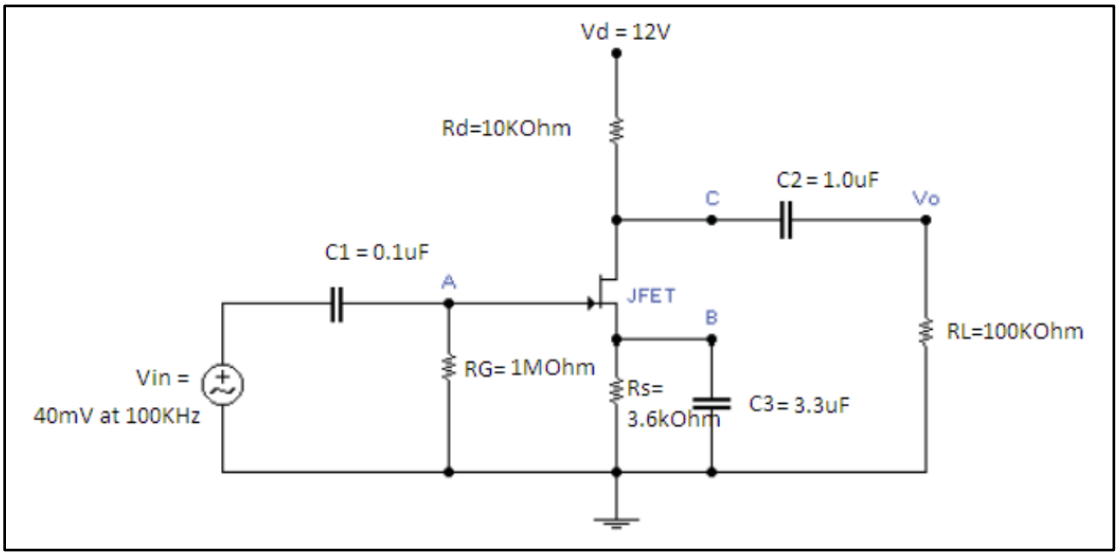


Figure #1: JFET Amplifier Schematic (Lab Manual)

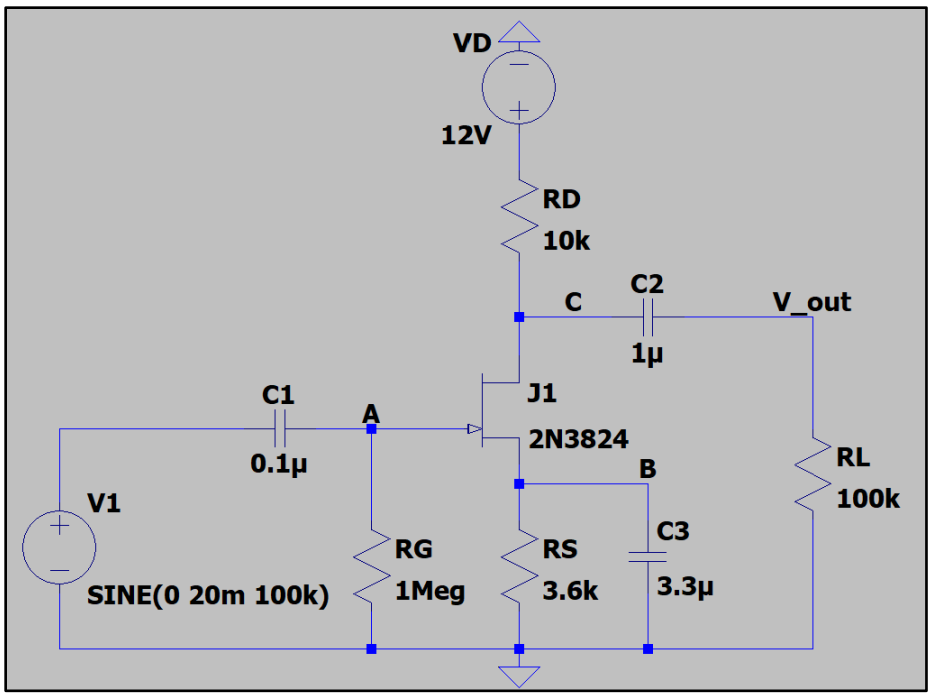


Figure #2: JFET Amplifier Circuit (LTSpice)

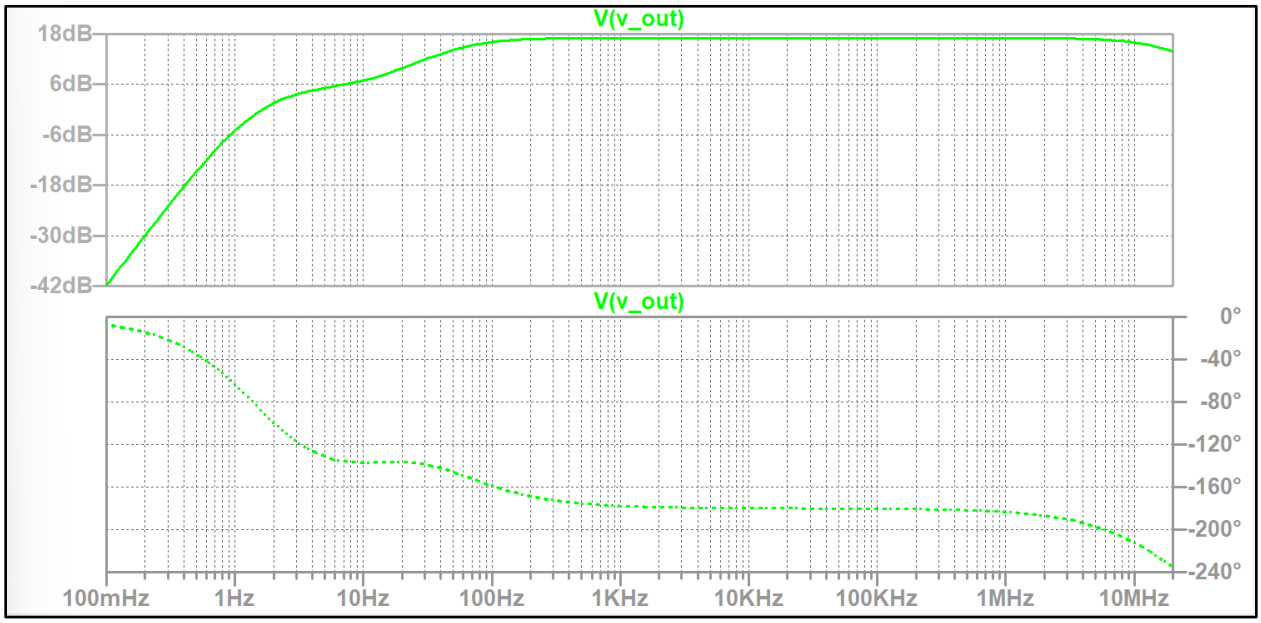


Figure #3: JFET Amplifier Bode Plot

	Value
Bandwidth	19.3 MHz
High Frequency Cutoff	19.3 MHz
Low Frequency Cutoff	46.9 Hz

Table #1: Important Bode Plot Values

Frequency (Hz)	V_{out} (mV)	Gain (dB)
30	154.94	11.76205
45	198.74	13.92439
60	219.48	14.78682
100	255.61	16.110466
200	276.4	16.789635
500	283.37	17.005928
1k	284.4	17.03745
10k	284.75	17.047975
100k	284.75	17.047962
500k	284.65	17.045081
1M	284.36	17.036147
1.5M	283.91	17.022409
2M	283.2	17.000765
3M	281.45	16.946799
4M	278.74	16.862811
5M	275.4	16.758145
7M	268.18	16.527211
10M	252.31	15.997369
11M	247.95	15.846186
12M	237.92	15.487421
15M	226.11	15.045178
16M	219.57	14.790256

Table #1: Simulations Results